

Function Reference

v1.2.1

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Function reference

Complete reference for the four worksheet functions exposed by the US Census Calc add-in. For setup (API key, build, install) see *INSTALL.md*; for a worked example spreadsheet see [demo/Census-Demo.ods](#) (generated by [tools/build_demo.py](#)).

All formulas below use Calc's **semicolon** argument separator. Arguments in `[brackets]` are optional and may be omitted entirely (trailing optional arguments can simply be left off the end of the formula).

Shared behavior

- **year** accepts either a number or a text cell (e.g. `2022` or `"2022"`).
- **dataset** is the API's dataset path, e.g. `"acs/acs5"` (ACS 5-year estimates), `"acs/acs1"` (ACS 1-year), or `"dec/pl"` (decennial redistricting data). Leading/trailing slashes are stripped automatically.
- **geography** maps to the API's `for=` clause, e.g. `"state:*` (every state), `"state:06"` (California only), or `"county:*` (every county — combine with `in_geography` to scope to one state).
- **in_geography** (where present) maps to the API's `in=` clause, e.g. `"state:06"`. Required whenever `geography` is below the state level (county, tract, block group, ...).
- **api_key** (where present) overrides the `CENSUS_API_KEY` environment variable for that one call. Type it literally or reference a cell, e.g. `$B$1`. **CENSUS_GET** and **CENSUS_VALUE** **require a key** — one or the other must be set, or the call raises a Calc error value (the API answers a keyless data request with a redirect to an HTML "missing key" page rather than data; see *INSTALL.md*). `CENSUS_VARLABEL` and `CENSUS_DATASETS` are metadata/discovery endpoints and never need a key, so they have no `api_key` argument.
- **Caching**: every distinct request URL is cached for the life of the LibreOffice session, so recalculating a sheet does not re-hit the API. Restart LibreOffice to clear the cache.
- **Errors**: invalid input, an API error response, or a network failure all surface as a genuine Calc error value in the cell (e.g. `Err:502`), never as a raw exception string or a silent wrong answer.
- **Array formulas**: `CENSUS_GET` and `CENSUS_DATASETS` can return more than one row. LibreOffice has no dynamic spill, so select an output range large enough for the result, type the formula, and confirm with **Ctrl+Shift+Enter** (or tick **Array** in the Function Wizard). A single-cell entry shows only the top-left value of the result.

CENSUS_GET

Runs a Census API data query and returns the full result table.

```
CENSUS_GET(year; dataset; variables; geography; [in_geography]; [api_key])
```

Argument	Type	Required	Description
<code>year</code>	number or text	yes	The data year, e.g. <code>2022</code> .

Argument	Type	Required	Description
dataset	text	yes	The dataset path, e.g. "acs/acs5" .
variables	text	yes	Comma-separated variable codes, e.g. "NAME,B01003_001E" . Mapped to the API's <code>get=</code> parameter.
geography	text	yes	The <code>for=</code> clause, e.g. "state:*" . May list several specific geographies, e.g. "state:06,36,48" .
in_geography	text	no	The <code>in=</code> clause, e.g. "state:06" .
api_key	text	yes, or via env var	Overrides <code>CENSUS_API_KEY</code> for this call. A key must come from one or the other.

Returns: a rectangular array. Row 1 is the column headers exactly as the API returns them (variable codes, then geography identifier columns such as `state` or `county`). Each subsequent row is one geography's values. Every cell is returned as **text**, matching the API's own JSON-array-of-arrays shape with minimal parsing — wrap numeric columns in `VALUE()` if you need them as numbers for arithmetic.

Errors: `dataset` / `variables` / `geography` empty, no API key available (neither the argument nor `CENSUS_API_KEY` set), an unknown dataset or variable, a malformed geography clause, or a network failure all raise a Calc error value. A query that legitimately returns zero rows (nothing matched) is also an error, since a table with no data is never useful in a sheet.

Examples:

```
=CENSUS_GET(2022; "acs/acs5"; "NAME,B01003_001E"; "state:*")
  -> 53 rows (header + 52 states/territories): name, total population

=CENSUS_GET(2022; "acs/acs5"; "NAME,B01003_001E"; "state:06,36,48")
  -> 4 rows: header + California, New York, Texas

=CENSUS_GET(2022; "acs/acs5"; "NAME,B01003_001E"; "county:*"; "state:06")
  -> 59 rows: header + every California county (in_geography scopes to CA)

=CENSUS_GET(2020; "dec/pl"; "NAME,P1_001N"; "state:*")
  -> 2020 decennial redistricting total population by state
```

CENSUS_VALUE

Returns a single scalar value for one variable at one specific geography.

```
CENSUS_VALUE(year; dataset; variable; geography; [in_geography]; [api_key])
```

Argument	Type	Required	Description
year	number or text	yes	The data year, e.g. 2022 .
dataset	text	yes	The dataset path, e.g. "acs/acs5" .
variable	text	yes	A single variable code, e.g. "B01003_001E" .

Argument	Type	Required	Description
geography	text	yes	The <code>for=</code> clause identifying one specific geography , e.g. <code>"state:06"</code> — not a wildcard (see Notes).
in_geography	text	no	The <code>in=</code> clause, e.g. <code>"state:06"</code> .
api_key	text	yes, or via env var	Overrides <code>CENSUS_API_KEY</code> for this call. A key must come from one or the other.

Returns: the requested value from the first (and normally only) data row — a **number** when it parses as one (most ACS estimates), **text** for string fields (e.g. `NAME`), or an **empty cell** when the API returns a null value for that variable/geography.

Errors: `dataset / variable / geography` empty, no API key available (neither the argument nor `CENSUS_API_KEY` set), no data row returned for the query, or the requested variable not present in the response header (e.g. a typo in the variable code) all raise a Calc error value.

Notes: internally this runs the same kind of query as `CENSUS_GET` with `variables` set to just `variable`, then reads the **first** data row. If `geography` resolves to more than one row (e.g. a wildcard like `"state:*"`), you get whichever row the API happened to return first — pass a specific geography instead, or use `CENSUS_GET` to pull every matching geography at once.

Examples:

```
=CENSUS_VALUE(2022; "acs/acs5"; "NAME"; "state:06")
  -> California

=CENSUS_VALUE(2022; "acs/acs5"; "B01003_001E"; "state:06")
  -> 39356104

=CENSUS_VALUE(2022; "acs/acs5"; "B19013_001E"; "county:075"; "state:06")
  -> median household income for San Francisco County, CA

=CENSUS_VALUE(2022; "acs/acs5"; "B01003_001E"; "state:06"; ""; "$B$1)
  -> same query, API key taken from cell B1
```

CENSUS_VARLABEL

Returns the human-readable label for a variable code, from the dataset's `variables.json` metadata.

```
CENSUS_VARLABEL(year; dataset; variable)
```

Argument	Type	Required	Description
year	number or text	yes	The data year, e.g. <code>2022</code> .
dataset	text	yes	The dataset path, e.g. <code>"acs/acs5"</code> .
variable	text	yes	The variable code to look up, e.g. <code>"B01003_001E"</code> .

Returns: the variable's `label` field (e.g. `"Estimate!!Total"`), or its `name` if the dataset has no label for it.

Errors: `dataset / variable` empty, or the variable code not found in that dataset/year, raises a Calc error value.

Notes: this endpoint is metadata, not data — it never needs an API key, so there is no `api_key` argument.

Examples:

```
=CENSUS_VARLABEL(2022; "acs/acs5"; "B01003_001E")
  -> Estimate!!Total

=CENSUS_VARLABEL(2022; "acs/acs5"; "B19013_001E")
  -> Estimate!!Median household income in the past 12 months (in 2022 inflation-adjusted dollars)
```

Pair it with `CENSUS_GET / CENSUS_VALUE` to build a self-documenting sheet: put variable codes in column A and `=CENSUS_VARLABEL(2022;"acs/acs5";A2)` in column B.

CENSUS_DATASETS

Lists the datasets available for a year, from the API's discovery endpoint.

```
CENSUS_DATASETS(year)
```

Argument	Type	Required	Description
year	number or text	yes	The data year, e.g. 2022 .

Returns: a two-column array. Row 1 is a `Dataset / Title` header row; each subsequent row is one available dataset's path (e.g. `"acs/acs5"` , joined from the API's `c_dataset` segments) and its title. A given year typically has ~100 datasets, so select a generous range (or a full column) when entering this as an array formula.

Errors: `year` empty, or no datasets listed for that year (e.g. a year far outside the API's coverage), raises a Calc error value.

Notes: this is the discovery endpoint, not a data query — it never needs an API key. Row order matches whatever order the API itself returns (not alphabetical); use `CENSUS_GET / CENSUS_VALUE` once you've found the dataset path you want.

Examples:

```
=CENSUS_DATASETS(2022)
  -> spills ~100 rows: (Dataset, Title) for every 2022 dataset

=CENSUS_DATASETS(2020)
  -> includes "dec/pl" ("Decennial Census: Redistricting Data (PL 94-171)")
```